

Credit Scorecard Reference

Score-to-decision lookup · feature attribution · measured performance
Credit Risk & Fraud Detection Platform · v1.1 · April 2026

RISK TIERS & DECISIONS

TIER A Super-prime 720+ PD < 5% Auto-approve	TIER B Prime 680-719 PD 5-10% Auto-approve	TIER C Near-prime 630-679 PD 10-20% Approve / refer	TIER D Subprime 580-629 PD 20-35% Refer / decline	TIER E Deep subprime 300-579 PD > 35% Decline
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SCORING METHODOLOGY — PDO SCALING

$$\text{Score} = 500 + 72.13 \times \ln(\text{odds})$$

where odds = (1 - PD) / PD

Reference score 600 at 4:1 odds (PD ≈ 20%)	PDO (points to double odds) 50 +50 pts → PD halves	Approval cutoff 638 PD = 28% threshold
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FEATURE-LEVEL SCORE ATTRIBUTION (TOP 10 BY INFORMATION VALUE)

#	Feature	IV	Direction	Max Impact
1	external_risk_estimate	0.247	Higher = better	±35 pts
2	loan_term_months *	0.190	36 mo = better	±25 pts
3	credit_score	0.156	Higher = better	±28 pts
4	pct_trades_never_delinquent	0.118	Higher = better	±22 pts
5	dti	0.094	Lower = better	±18 pts
6	num_high_utilization_trades	0.087	Lower = better	±16 pts
7	score_x_product *	0.082	Mixed	±15 pts
8	ltv_x_product *	0.071	Lower = better	±13 pts
9	num_derogatory_marks	0.064	Lower = better	±12 pts
10	annual_income	0.058	Higher = better	±11 pts

* Excluded from v1.1 segmented per-product models — encodes product-type signal rather than credit risk (see MODEL_CARD §6).

MEASURED TIER PERFORMANCE — OOT TEST SET

Tier	Loans	Avg Score	Predicted PD	Actual DR	Gap
C — Near-prime	39,257	640	5.98%	8.67%	-2.69 pp
D — Subprime	133,892	608	17.00%	23.48%	-6.48 pp

Negative gap on Tier D reflects the loan-term-months product proxy (MODEL_CARD §6). The v1.1 segmented Unsecured model achieves AUC 0.7208 vs 0.6678 unified, materially closing this gap.

ADVERSE ACTION REASON CODES (ECOA-COMPLIANT)

Code	Description	Mapped From
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AA-01	Insufficient credit history	months_since_oldest_trade low
AA-02	Too many recent inquiries	num_inquiries_last_6m high
AA-03	Debt-to-income ratio too high	dti high
AA-04	Insufficient income	annual_income low
AA-05	Recent derogatory marks	num_derogatory_marks ≥ 1
AA-06	High credit utilization	num_high_utilization_trades high
AA-07	Credit score below threshold	credit_score low
AA-PF	Hard policy failure	Policy gate (specific rule cited)